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# Evaluation of antibacterial activity of leaf and seed extracts of *Ocimum sanctum*

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**Abstract-** Antibacterial evaluation of *Ocimum sanctum* of family Lamiaceae has been done for their methanolic and aqueous extract of leaf and seed against pathogenic strain of gram negative bacteria *Escherichia coli, Pseudomonas aeruginosa* and gram positive bacteria *Staphylococcus aureus*. Methanolic extract of both the seed and leaf has pronounced antibacterial activity then aqueous extracts. Methanolic seed extract showed highest antibacterial activity against *Staphylococcus aureus* (22 mm) in comparison to *E.coli* (18 mm), *Pseudomonas aeruginosa* (17mm) and standard antibiotic drug Chloramphenicol (20 mm). Thus methanolic extract of *O. sanctum* is more effective at 30mg/ml concentration in treating the diseases than aqueous extract. It may be the best alternatives of allopathic drug to cure bacterial diseases.

**Key words: Antimicrobial activity, Chloramphenicol, Disc diffusion method, *Ocimum sanctum***

## INTRODUCTION

Nature has bestowed man with innumerable resources. Plant extracts are used to cure microbial diseases since ancient times in traditional medicinal system. Capacity of utilizing the mass of the medicinal plant for the treatment and antimicrobial effect of phytochemicals.[1] Approximate 80% of the world population depended on the use of long-established medicine which is predominantly based on plant material.[2] Medicinal plants are widely used in drug as a source of natural products.[3] In the present time, a number of sensitive microorganisms to human beings and plants become resistance due to indiscriminate use of antibiotics. This will create a big problem in controlling the growth of pathogenic bacterium to be treated with antibiotics in future. As a result, scientists are forced to search for new antimicrobial compound from various parts of medicinal plants. In such a situation medicinal plants rich in antimicrobial substances could be used to kill the microorganisms as an alternative medicine instead of synthetic medicine and antibiotics.[4] Antimicrobial activity of some phytochemicals have been looked over and the possibility of using them in the development of new antimicrobial drugs also been reported.[5] Some of these phytochemicals of medicinal importance present in *Ocimum sanctum* have already been identified.[6] Several of these phytochemicals have been exposed to possess functional biological activities belonging mainly to phenolic, flavonoid, and carotenoid compounds.[7]

*Ocimum sanctum* has been analysed for their antimicrobial activity, but no another species have been investigated in such a way.[8] All over the old world and

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extremely in the tropics *Ocimum sanctum* known as Tulsi or Holy basil is till abundantly for religious and medicinal purposes. It is mostly used in Ayurveda and across the wide region of south-east Asia is widely known as a medicinal plant as well as an herbal tea. *Ocimum sanctum* belongs to family Labiaceae. It is found all over tropical and semitropical zone of India and other Asian countries.[9,10] *Ocimum sanctum* plant role paly as insecticide, nematicide, fungicide and antimicrobial compound also has been documented.[11] Extracts of these chemicals from *Ocimum sanctum* plant retain useful pharmacological applications. Although, reported on the antimicrobial activity of *Ocimum sanctum* extracts against pathogen.

In our present work, we extended the study to screen the antimicrobial activity of methanolic and aqueous extracts of the leaves and seeds of Ocimum sanctum against three different gram positive and gram negative bacterial strains.

## MATERIAL & METHODS

**Plant material collection : -**

Fresh plant material such as leaves and seeds, were collected from the Chhau village of Jhunjhunu district, Rajasthan from the natural environment. Plant materials were washed, dried at room temperature, crushed to fine powder and stored until future use.

**Preparation of Extract : -**

**Aqueous extract -** Samples of 10g, 20g and 30g were placed in a conical flask immersed in 100 ml of distilled water plugged with cotton and kept on a rotary shaker for 3 hrs. The extracts were filtered by watchman no. 1 filter paper, centrifuged at 5000 rpm for 15 min. and the supernatant was clam and the solvent was evaporated to make volume one fourth of its original volume. The extracts were stored at 4°C in airtight bottles.

**Methanol extract -** Air dried samples powder of 10g, 20g and 30g were placed in a conical flask immersed in 100 ml of methanol plugged with cotton and kept on a rotary shaker for 3 hrs. The extracts were filtered by watchman no. 1 filter paper, centrifuged at 5000 rpm for 15 min. and the supernatant was clam and the solvent was evaporated to make volume one fourth of its original volume. The extracts were stored at 4°C in airtight bottles.

**Antimicrobial activity: -**

Disc diffusion method -*In vitro* antibacterial activity of the crude aqueous and methanol extracts were studied against gram positive (*Staphylococcus aureus*), gram negative (*Escherichia coli, Pseudomonas aeruginosa*) bacterial strains by the disc diffusion method. The antimicrobial activity of methanol and aqueous extracts were carried out.[11] Each plant extracts was tested at three different concentrations (10g, 20g and 30g per 100 ml) to see their inhibitory effects against microbial pathogens. Sterile paper disc (6 mm in diameter) prepared from whatman no. 1 filter paper. Both aqueous and methanol extracts of the leaves and seeds of *Ocimum sanctum* exhibited moderate antimicrobial activity with the all tested strain at (10g, 20g, 30g per 100ml) concentrations when compared with the standard chloramphenicol (1%), containing solution transferred on the inoculated agar. The inoculated plates were incubated at 37°C for 24 hrs. The antimicrobial activity was assay by measuring the diameter of the inhibition zone for the three test micro-organisms.

## RESULT & DISCUSSION

The result of the antibacterial activities of the methanolic and aqueous leaf and seed extract of *Ocimum sanctum* were compared with the standard antibiotic chloramphenicol (1%) shown in table 1. The methanolic leaf extract at 30mg/ml concentration showed (20 mm) zone of inhibition against *Staphylococcus aureus* which is equal to standard chloramphenicol (20 mm). At the same concentration *E.coli* showed 17 mm inhibitory zone and *Pseudomonas aeruginosa* showed 16mm zone of inhibition, which is equivalent to antimicrobial effect of synthetic drugs. *Ocimum* leaf extract showed maximum zone of inhibition 16 mm against *Pseudomonas aeruginos* compared to chloramphenicol (15 mm). By analysing the result we observed that methanolic leaf extract of *Ocimum sanctum* at 30mg/100ml concentration showed similar results to standard synthetic drugs against *E. coli* (17 mm). The seed extract at 30mg/100ml concentration showed better results (22mm, 18mm and 17mm) than standard (20mm, 17mm and 15mm) synthetic drugs against all bacteria *Staphylococcus aureus, Pseudomonas aeruginos* and *E.coli.* Methanolic extract showed higher antibiotic activity in comparison to aqueous extract. 30g/100ml concentration showed best results in comparison to 10g/100ml, 20g/100ml in both plant materials for all type of bacteria.

**Table 1- Observation table of antimicrobial activities of methanol and aqueous extracts of *Ocimum sanctum***

| Name of the Plant material | Name of the strain | Quantity of plant material g/100ml | Zone of inhibition (mm) | | |
|---|---|---|---|---|---|
| | | | Methanol (mm) | Aqueous (mm) | Synthetic drug (chloramphenicol) |
| Leaves | *Staphylococcus aureus* | 10 | 11 | 9 | 20 |
| | | 20 | 16 | 11 | |
| | | 30 | 20 | 14 | |
| | *Escherichia coli* | 10 | 12 | 9 | 17 |
| | | 20 | 13 | 12 | |
| | | 30 | 17 | 14 | |
| | *Pseudomonas aeruginosa* | 10 | 10 | 9 | 15 |
| | | 20 | 12 | 11 | |
| | | 30 | 16 | 13 | |
| Seed | *Staphylococcus aureus* | 10 | 14 | 11 | 20 |
| | | 20 | 17 | 13 | |
| | | 30 | 22 | 16 | |
| | *Escherichia coli* | 10 | 13 | 10 | 17 |
| | | 20 | 15 | 12 | |
| | | 30 | 18 | 15 | |
| | *Pseudomonas aeruginosa* | 10 | 12 | 10 | 15 |
| | | 20 | 14 | 11 | |
| | | 30 | 17 | 14 | |

**Figure- Antimicrobial activity was expressed in terms of the diameter of zone of growth inhibition**

1. ***Ocimum sanctum* leaves methanolic extracts : -**




**1.1- *Staphylococcus aureus***




**1.2- *Escherichia coli***




**1.3- *Pseudomonas aeruginosa***

2. ***Ocimum sanctum* seeds methanolic extracts : -**




**2.1- *Staphylococcus aureus***




**2.2- *Escherichia coli***




**2.3- *Pseudomonas aeruginosa***

## CONCLUSION

Plants are the biggest source of medicine. The present study exhibited the antibacterial effect of aqueous and methanolic extract of *Ocimum sanctum* against *Staphylococcus aureus, Pseudomonas aeruginos* and *E.coli*. The methanolic extract is so powerful then aqueous extract, it may be due to the presence of alkaloids, Steroids, Saponins, Flavanoids in methanol extract of *Ocimum sanctum* leaf. The plant extract are safe, easy, effective and practical solutions to every day ailments and do not show any toxic and side effects.

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